TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

Bachelor of Artificial Intelligence and Robotics Engineering

AI.403

Robot Learning

COURSE OVERVIEW

Robots that improve with experience: imitation, reinforcement, meta-learning, and lifelong learning on hardware.

ROBOT LEARNING

1

SESSION

60 MINUTES

ORIENTATION: ROBOTS THAT IMPROVE

KEY TOPICS

- Robot learning vs. general ML
- Learning on physical robots
- Sample efficiency
- Course roadmap

TEACHING FOCUS

Let robots get better with practice.

KEY TAKEAWAY Experience is the best teacher for robots.

ROBOT LEARNING

2

SESSION

60 MINUTES

IMITATION LEARNING

KEY TOPICS

- Learning from demonstration
- Behavior cloning
- Inverse reinforcement learning
- DAgger

TEACHING FOCUS

Teach robots by showing them.

KEY TAKEAWAY Demonstration is an efficient teacher.

ROBOT LEARNING

3

SESSION

60 MINUTES

REINFORCEMENT LEARNING ON ROBOTS

KEY TOPICS

- RL on hardware
- Safe exploration
- Sim-to-real
- Reward shaping

TEACHING FOCUS

Learn policies through interaction.

KEY TAKEAWAY Hardware teaches grounding.

4

SESSION

60 MINUTES

LIFELONG & META-LEARNING

KEY TOPICS

- Continual learning
- Catastrophic forgetting
- Meta-learning
- Few-shot robot learning

TEACHING FOCUS

Learn to learn, and never stop.

▲ **KEY TAKEAWAY** Lifelong learners adapt forever.

5

SESSION

60 MINUTES

SKILL GENERALIZATION

KEY TOPICS

- Generalizing across tasks
- Transfer learning
- Domain adaptation
- Reusable skills

TEACHING FOCUS

Apply one skill to many situations.

KEY TAKEAWAY Generalization is the mark of real learning.

ROBOT LEARNING

6

SESSION

60 MINUTES

ROBOT LEARNING & REVIEW

KEY TOPICS

- Learning robot demo
- Course recap and assessment
- Toward AI ethics (AI.404)
- Skill showcase

TEACHING FOCUS

Demonstrate a robot learning a skill.

KEY TAKEAWAY Learning robots grow with use.